6 High-Level Design (Architecture)

Chapter 8. The rule now reverses: a high-level design that does not say where things run, what they are built from and what that choice cost has not been done.

6.1 Architectural drivers: quality-attribute scenarios

QAS	Quality attribute	Stimulus	Artifact	Response	Response measure	From
QAS-01	Performanc e	A logged in student with 200 application records opens the dashboard	Dashboard and Data Access Componen t	The system shows the total count, the count in each of the 4 statuses, and reminders for the next 7 days	95% of 100 dashboard loads complete within 2 seconds	QR–01
QAS-02	Security (authorizati on)	A logged in student sends a request to view, edit, delete, or download another student's application record	Authentica tion Componen t (AccessCh eck) and every componen t that returns or changes records	The system refuses the request and nothing is changed or given out	100% of at least 20 cross-account requests are denied	QR-02
QAS-03	Security (confidentia lity)	An attacker obtains a full copy of the stored account data	Authentica tion componen t and the stored account data	No password can be read from the stored data	0 of the stored values match any entered password.	QR-03
QAS-04	Usability	A first time student logs in and wants to record an internship	User interface and application managem ent	The student saves the record and sees it in the list with status applied without help.	At least 4 of 5 first time users save their first record within 3 minutes of login	QR-04

Constraints: CON-1 means InterTrack owns registration, login and password storage, so an authentication component is required as QAS-03 applies to our data. CON-02 leaves the file storage service as the only external system. CON-03 means reminders are computed inside the system and no messaging service is needed.

6.2 Architectural style and alternatives

Style considered	Drivers it serves	What it costs	Verdict
Layered: Browser layer, server layer, data access layer	QAS-02: Every request enters through one application layer, where AccessCheck runs before data is read. QAS-04: One server and one database keep the dashboard to a single path	Each request passes through every layer, which adds some pass through code and processing time. The single application server is a single failure, and changes redeploy the entire server.	Chosen
Microservices: one service per component	Independent deployment and scaling of each service	6+ services to build, deploy with a 5 person team in one semester.	Rejected
Two tiered Client server	Fastest to build, fewest components, low latency for QAS-01	The browser is untrusted, so QAS-02 would entirely require the database rules written correctly. Many rules would run in the client and could be bypassed.	Rejected

We chose the layered client server because our two MUST quality scenario are both security related. QAS-02 requires 100% of all cross account requests to be denied, and the layered style gives a single place that every requests must pass before it gets to access data. That makes the access check impossible to skip from the browser, which is the primary weakness of the two tiered client server option.

6.3 Component view

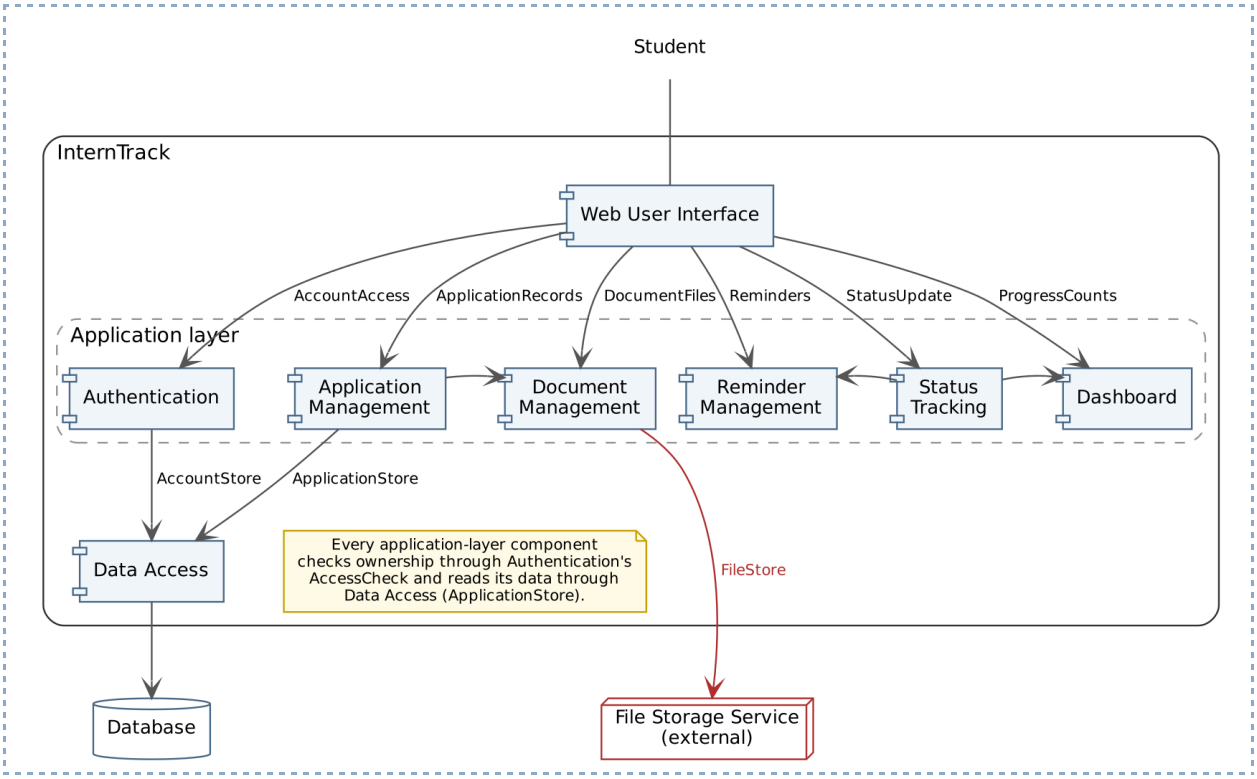


Figure 6.3 Component diagram: components, provided and required interfaces, external systems

Component	Responsibility (one sentence)	Provides	Requires	Answers to (UC / QAS / ADR)
Web User Interface	Shows register and login pages, application list, add and edit form, status history, reminders and dashboard	Pages to the student	All 6 interfaces	UC-01-UC-05, QAS-04, CON-04, ADR-01
Authentication	Registers accounts, hashes and verifies passwords, decides whether a requested record belongs to the student	AccountAccess, AccessCheck	AccountStore	UC-05, QAS-02, QAS-03, CON-01
Application Management	Creates, shows, edits, and deletes application records and interview dates	ApplicationRecords	AccessCheck, ApplicationStore, DocumentFiles	UC-01, UC-04, QAS-02, QAS-04

Component	Responsibility (one sentence)	Provides	Requires	Answers to (UC / QAS / ADR)
Status Tracking	Changes an application's status, denies changing to the same status, and records previous statuses.	StatusUpdate	AccessCheck, ApplicationStore, Reminders, ProgressCounts	UC-02, QAS-02
Document Management	Accepts PDFs of at most 10MB and stores and retrieves them through the file storage	DocumentFiles	AccessCheck, ApplicationStore, FileStore	UC-01, CON–03
Reminder Management	List student's deadlines and interviews in the next 7 days	Reminders	AccessCheck, ApplicationStore	UC-03, ADR-03, CON-03
Dashboard	Counts the student's applications	ProgressCounts	AccessCheck, ApplicationStore	UC-03, QAS-01
Data Access	Only component that reads and writes to the database	AccountStore, ApplicationStore	Database	All UCs, QAS-01, QAS-02, ADR-01
File Storage Service	Holds the uploaded PDF files	FileStore	-	FR-12, ADR-02

6.4 Deployment view

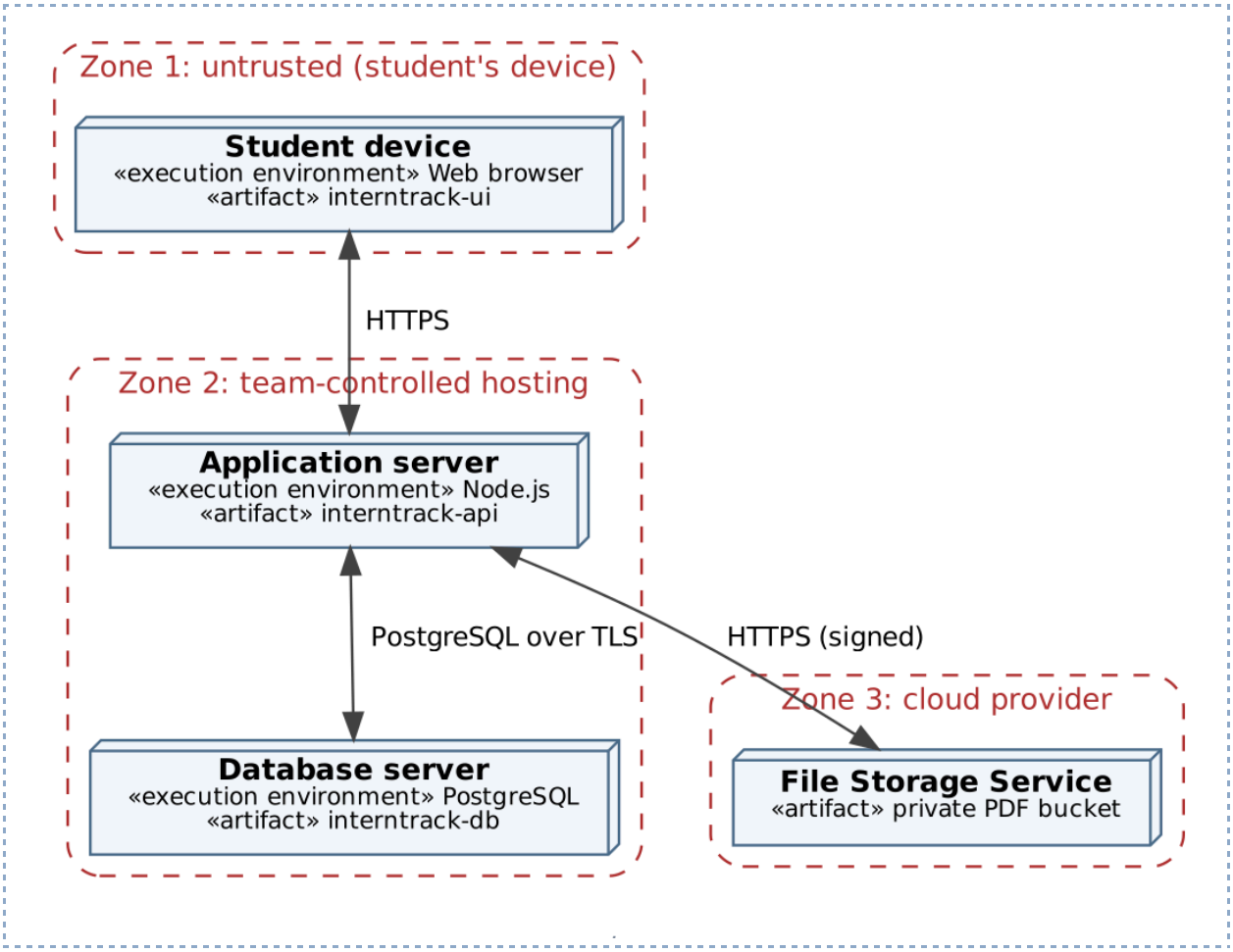


Figure 6.4 UML deployment diagram: nodes, execution environments, deployed artifacts, communication paths with protocols, trust boundaries

Node	What runs on it (artifacts, components)	Communicates with (protocol)	Trust zone
Student device (phone/computer)	Artifact interntrack-ui	Application server	Zone 1, untrusted. Everything from here is validated on server
Application server	Artifact interntrack-api, which is authentication application management, status tracking, document management, reminder management, dashboard, data access	Browser, database server, file storage service	Zone 2, team controlled

Node	What runs on it (artifacts, components)	Communicates with (protocol)	Trust zone
Database server	Artifact interntrack-db: accounts, applications, interviews, status changes, document details	Application server	Zone 2, team controlled
File Storage Service	PDF files	Application server	Zone 3